Plant nutrition

Mark scheme and worked solutions for 20 questions: 10 multiple choice, 10 structured.

1 · Part 1

Core · Photosynthesis · [1]

Answer: C

Explanation

Photosynthesis is the process by which plants synthesise carbohydrates from raw materials (carbon dioxide and water) using energy from light. Option A describes respiration. Option B is wrong because plants make their own carbohydrates; roots absorb water and mineral ions, not carbohydrates. Option D reverses the gases: in photosynthesis carbon dioxide is taken in and oxygen is given out.

2 · Part 1

Core · Photosynthesis · [1]

Answer: A

Explanation

Chlorophyll is found in chloroplasts and transfers energy from light into energy in chemicals, which is used to make carbohydrates. Option B gives the wrong organelle. Option C is wrong because chlorophyll does not absorb carbon dioxide; carbon dioxide diffuses into cells. Option D describes respiration in mitochondria.

3 · Part 1

Core · Photosynthesis · [1]

Answer: D

Explanation

Plants transport carbohydrate in the phloem as sucrose, and they build cell walls from cellulose. Glucose and starch are not the transported form, and starch is an energy store, not a cell wall material, so options A, B and C are each wrong in at least one column.

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4 · Part 1 Core Photosynthesis [1]

Answer: B

Worked solution

Link each mineral ion to the substance it is used to make: nitrate ions make amino acids; magnesium ions make chlorophyll. Then follow the chain: less chlorophyll leads to less light energy transferred, less photosynthesis, less glucose, less growth.

5 · Part 1 Core Photosynthesis [1]

Answer: A

Worked solution

Iodine solution turns blue-black only where starch is present. Starch is made by photosynthesis, which needs chlorophyll and light. Check each region for both: only P has both.

6 · Part 1 Core Photosynthesis [1]

Answer: C

Explanation

In bright light the pondweed photosynthesises faster than it respires, so it takes in carbon dioxide and the indicator turns purple. In the dark (foil) it only respires, so carbon dioxide rises and the indicator turns yellow. Tube 3 has no living organism, so the carbon dioxide concentration does not change and it stays orange-red. Option A reverses tubes 1 and 2; option B assumes the plant photosynthesises in the dark; option D wrongly assumes light alone changes the indicator.

7 · Part 1 Supplement Photosynthesis [1]

Answer: D

Worked solution

Multiply every part of the equation by 2: $12CO_2 + 12H_2O \rightarrow 2C_6H_{12}O_6 + 12O_2$. Check: carbon atoms 12 on each side; oxygen atoms 24 + 12 = 36 on the left and 12 + 24 = 36 on the right.

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8 · Part 1 Supplement Photosynthesis [1]

Answer: B

Worked solution

To find a limiting factor, look for the factor that, when increased, increases the rate. Compare the two columns at the same light intensity (tests carbon dioxide), then read down a column (tests light intensity).

9 · Part 1 Core Leaf structure [1]

Answer: C

Explanation

Thin leaves give a short diffusion distance, so carbon dioxide reaches the mesophyll cells quickly (and light reaches all the cells). Option A is wrong because water loss is not an adaptation for photosynthesis. Thinness does not change how many chloroplasts each cell holds (option B). Being thin does not reduce surface area (option D); leaves have a large surface area.

10 · Part 1 Core Leaf structure [1]

Answer: A

Explanation

The palisade mesophyll is just below the upper epidermis, so its cells receive the most light for photosynthesis. Air spaces in the spongy mesophyll allow gases to diffuse; they do not store water (option B). Upper epidermis cells are transparent with few or no chloroplasts, so they let light pass through to the mesophyll (option C). Xylem carries water and mineral ions into the leaf; sucrose is carried by the phloem (option D).

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11 · Part 1 Core Photosynthesis [6]

Mark scheme

(a) **[2]**

- plants make / synthesise carbohydrates (1)
- from raw materials / from carbon dioxide and water, using energy from light (1)

ALLOW glucose / sugar / starch for carbohydrates

REJECT "making energy" or "producing light energy"

(b) **[2]**

- carbon dioxide + water → (1)
- glucose + oxygen (1)

IGNORE light and chlorophyll written above or beside the arrow

ALLOW the correct balanced symbol equation

REJECT any extra product such as energy

(c) **[1]**

- chloroplast(s) (1)

(d) **[1]**

- transfers energy from light into energy in chemicals (for making carbohydrates) (1)

ALLOW absorbs (energy from) light and transfers it to chemical energy

REJECT "makes energy" / "produces light"

Worked solution

Photosynthesis is the process by which plants synthesise carbohydrates from raw materials using energy from light. The raw materials are carbon dioxide (from the air) and water (from the soil), and the products are glucose and oxygen: carbon dioxide + water → glucose + oxygen, in the presence of light and chlorophyll. Chlorophyll is a green pigment found in chloroplasts. Its job is to transfer energy from light into energy in chemicals, which the plant uses to make carbohydrates. Chlorophyll does not create energy; it transfers it from one form to another.

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12 · Part 1 Core Photosynthesis [6]

Mark scheme

(a) **[1]**

- for transport (in the phloem) to other parts of the tree (1)

ALLOW sucrose is the form in which carbohydrate is transported in the phloem

(b) **[2]**

- nectar attracts insects (1)
- insects transfer pollen (between flowers) / insects pollinate the flowers (1)

(c) **[2]**

- (energy comes from) starch stored (in the stem / roots / buds) during the summer (1)
- starch is broken down to glucose, which is used in respiration to release / provide energy (1)

ALLOW sugar for glucose

REJECT "starch provides energy" without reference to respiration for the second mark

REJECT photosynthesis in the new leaves

(d) **[1]**

- cellulose (1)

Worked solution

Glucose made in the leaves has several uses. It is changed into sucrose for transport in the phloem to other parts of the plant. Some is made into nectar in flowers, which attracts insects; the insects carry pollen from flower to flower, so the flowers are pollinated. Extra carbohydrate is stored as starch, an energy store. When the tree has no leaves it cannot photosynthesise, so in spring it uses its starch stores: the starch is broken down to glucose, and the glucose is used in respiration to provide the energy for growing new leaves. New cell walls are built from cellulose, which is also made from glucose.

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13 · Part 1 Core Photosynthesis [7]

Mark scheme

(a) **[2]**

- increase = 4.8 − 0.5 = 4.3 (g) and (4.3 ÷ 0.5) × 100 (1)
- 860 (%) (1)

ALLOW correct answer with no working for 2 marks

(b) **[3]**

- magnesium ions are needed to make chlorophyll (1)
- less chlorophyll so the leaves are yellow / less green (1)
- less energy from light is transferred, so less photosynthesis / less glucose made for growth (1)

(c) **[2]**

- nitrate ions are needed to make amino acids (1)
- amino acids are needed to make proteins (for new cells / growth) (1)

REJECT nitrate ions are needed to make chlorophyll

Worked solution

(a) Percentage increase = (final − initial) ÷ initial × 100. The increase is 4.8 − 0.5 = 4.3 g, so the percentage increase is 4.3 ÷ 0.5 × 100 = 860%. Check by working backwards: 0.5 g increased by 860% is 0.5 + (8.6 × 0.5) = 0.5 + 4.3 = 4.8 g.

(b) Magnesium ions are used to make chlorophyll, the green pigment. Without them the plant makes little chlorophyll, so the leaves look yellow. With less chlorophyll, less energy from light can be transferred for photosynthesis, so less glucose is made and there is less carbohydrate for growth, giving a lower mass.

(c) Nitrate ions are used to make amino acids, and amino acids are joined to make proteins. Proteins are needed to make new cells, so without nitrate ions growth is very poor.

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14 · Part 1 Core Photosynthesis [7]

Mark scheme

(a) **[2]**

- to remove / use up the starch already in the leaves / to destarch the leaves (1)
- so that any starch found at the end must have been made during the investigation (1)

ALLOW so the starch test is a fair test of photosynthesis during the 6 hours

(b) **[1]**

- to remove the chlorophyll / green colour so that the colour change with iodine can be seen (1)

IGNORE to soften the leaf

(c) **[3]**

- plant X: (orange-)brown / stays the colour of iodine solution (1)
- plant Y: blue-black (1)
- (plant X) had no carbon dioxide so it could not photosynthesise, so no starch was made (1)

ALLOW yellow-brown or orange for plant X

ALLOW black or dark blue for plant Y

(d) **[1]**

- it is a control, to show that the result for plant X is caused by the lack of carbon dioxide (and not by another factor) (1)

ALLOW to compare with plant X / to show that starch is made when carbon dioxide is present

Worked solution

(a) Plants store starch, so before the investigation the leaves already contain some. Keeping the plants in the dark for 48 hours means they cannot photosynthesise and they use up their starch. Any starch found at the end must therefore have been made during the 6 hours in the light.

(b) Hot ethanol removes the chlorophyll, which would otherwise hide the colour change. Boiling water first kills and softens the leaf.

(c) Iodine solution is orange-brown and turns blue-black when starch is present. Soda lime removed the carbon dioxide from the bag around plant X, so plant X could not photosynthesise and made no starch: its leaf stays orange-brown. Plant Y had carbon dioxide and light, so it photosynthesised and made starch: its leaf turns blue-black.

(d) Plant Y is a control. It is treated in exactly the same way except that it has carbon dioxide, so any difference between X and Y must be due to carbon dioxide.

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15 · Part 1 Core Photosynthesis [7]

Mark scheme

(a) **[1]**

- 22 (1)

(b) **[2]**

- as the distance increases, the number of bubbles per minute decreases (1)
- correct data quote with units, e.g. mean 48 bubbles per minute at 10 cm falling to 7 bubbles per minute at 50 cm (1)

ALLOW the decrease is greatest at shorter distances / the decrease gets smaller as distance increases

(c) **[2]**

- light intensity decreases (as the lamp moves further away) (1)
- (so) less energy from light (is available) for photosynthesis / rate of photosynthesis decreases, so less oxygen is made (1)

ALLOW light becomes the limiting factor

(d) **[1]**

- oxygen (1)

(e) **[1]**

- put a (glass) tank / beaker of water between the lamp and the pondweed (1)

ALLOW use an LED lamp / a lamp that gives off little heat

ALLOW measure the temperature of the water with a thermometer and keep it constant (e.g. using a water bath)

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Worked solution

(a) Mean = (19 + 24 + 23) ÷ 3 = 66 ÷ 3 = 22 bubbles per minute. Check: 22 is between the lowest (19) and highest (24) values, as a mean must be.

(b) The further away the lamp, the fewer bubbles per minute: the mean fell from 48 at 10 cm to 7 at 50 cm. The biggest drop is between 10 cm and 20 cm.

(c) Moving the lamp further away lowers the light intensity reaching the pondweed. Less energy from light is available, so the rate of photosynthesis decreases and less oxygen is released, giving fewer bubbles.

(d) Photosynthesis releases oxygen, so the bubbles are oxygen.

(e) Temperature also affects the rate of photosynthesis, so it must be kept constant. A tank of water between the lamp and the pondweed absorbs heat but lets light through.

16 · Part 1 Supplement Photosynthesis [8]

Mark scheme

(a) **[2]**

- volume (of oxygen) increases as temperature increases from 10 °C to 35 °C (1)
- maximum / optimum at 35 °C and then decreases (to 45 °C) (1)

(b) **[2]**

- (3.0 − 0.8) ÷ 3.0 × 100 / 2.2 ÷ 3.0 × 100 (1)
- 73 (%) (1)

ALLOW correct answer with no working for 2 marks

REJECT 73.3 for the second mark (not to the nearest whole number)

(c) **[2]**

- enzymes (that control photosynthesis) are denatured (1)
- the active site changes shape so the substrate no longer fits / fewer enzyme-substrate complexes form (1)

REJECT enzymes are killed

(d) **[2]**

- temperature (1)
- increasing the temperature (above 20 °C) increases the volume of oxygen / rate, and light intensity and carbon dioxide concentration are already high (1)

Worked solution

(a) As temperature rises from 10 °C to 35 °C the volume of oxygen rises, from 0.6 cm^3 to 3.0 cm^3. The maximum is at 35 °C. Above this the volume falls to 0.8 cm^3 at 45 °C.

(b) Percentage decrease = (original − new) ÷ original × 100 = (3.0 − 0.8) ÷ 3.0 × 100 = 2.2 ÷ 3.0 × 100 = 73.3…, which is 73% to the nearest whole number. Check: 73% of 3.0 is about 2.2, and 3.0 − 2.2 = 0.8.

(c) The reactions of photosynthesis are controlled by enzymes. Above the optimum temperature the enzymes are denatured: the shape of the active site changes, so the substrate no longer fits and fewer enzyme-substrate complexes form. The rate of photosynthesis falls, so less oxygen is produced.

(d) A limiting factor is the factor in shortest supply, which, if increased, would increase the rate. At 20 °C, light and carbon dioxide are already high, and raising the temperature increases the rate, so temperature is limiting.

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17 · Part 1 Core Photosynthesis [6]

Mark scheme

(a) **[2]**

- no light so no photosynthesis (1)
- the plant respires, releasing carbon dioxide, so the carbon dioxide concentration increases (1)

(b) **[2]**

- the rate of photosynthesis is greater than the rate of respiration (1)
- (so) more carbon dioxide is taken in than released / carbon dioxide is used up, so the carbon dioxide concentration decreases (1)

ALLOW "photosynthesis uses up carbon dioxide faster than respiration releases it" for both marks

(c) **[1]**

- the rate of photosynthesis equals the rate of respiration / carbon dioxide used in photosynthesis equals carbon dioxide released in respiration (1)

(d) **[1]**

- (control) to show that the colour change is caused by the (aquatic) plant / to show that light alone does not change the colour of the indicator (1)

Worked solution

(a) Tube 3 is in the dark, so the plant cannot photosynthesise. It still respires, releasing carbon dioxide into the solution, so the carbon dioxide concentration rises above that in air and the indicator turns yellow.

(b) In bright light the plant photosynthesises faster than it respires. It takes in more carbon dioxide than it releases, so the carbon dioxide concentration falls below that in air and the indicator turns purple.

(c) The cloth lets only dim light through. In dim light the plant takes in carbon dioxide for photosynthesis at the same rate as it releases carbon dioxide by respiration, so there is no overall change.

(d) Tube 4 has no plant. Because it stays orange-red, it shows that the colour changes in the other tubes are caused by the plant and not by the light or the tube.

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18 · Part 1 Supplement Photosynthesis [7]

Mark scheme

(a) **[2]**

- $6CO_2 + 6H_2O \rightarrow$ (1)
- $C_6H_{12}O_6 + 6O_2$ (1)

ALLOW CO2, H2O, C6H12O6, O2 without subscripts

REJECT a word equation

(b) **[2]**

- carbon dioxide concentration (1)
- the rate increases when the carbon dioxide concentration increases / the rate is the same at both light intensities (so light is not limiting) (1)

ALLOW a data quote for the second mark, e.g. the rate doubles from 6 to 12 at both light intensities

(c) **[2]**

- light intensity is limiting (1)
- increasing carbon dioxide has no effect, but at high light intensity the rate is higher at the same carbon dioxide concentration (1)

ALLOW a data quote for the second mark, e.g. at 0.06% the rate is 16 at low light but 33 at high light

(d) **[1]**

- temperature / light intensity (1)

REJECT carbon dioxide concentration

IGNORE water availability

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Worked solution

(a) The balanced equation is $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$.

(b) Between 0.01% and 0.02% carbon dioxide, the rate doubles from 6 to 12 arbitrary units at both light intensities. Because the rate is the same at both light intensities, light is not limiting; because increasing carbon dioxide increases the rate, carbon dioxide concentration is the limiting factor.

(c) At low light intensity, adding more carbon dioxide above 0.04% makes no difference. At the same carbon dioxide concentrations the rate is much higher in high light, so the low light intensity is what is holding the rate back.

(d) At high light the rate levels off at 38 between 0.08% and 0.10%, so carbon dioxide is no longer limiting. Another factor must be limiting; temperature (fixed at 25 °C) or light intensity could be.

19 · Part 1 Supplement Photosynthesis [7]

Mark scheme

(a) **[2]**

- temperature (1)
- the rate increases from 20 to 45 when the temperature increases from 8 °C to 18 °C, while light intensity and carbon dioxide concentration stay the same (1)

(b) **[3]**

- the heater raises the temperature (from 8 °C to 18 °C), so temperature is no longer limiting (1)
- the heater raises the carbon dioxide concentration, so carbon dioxide is no longer limiting / the rate rises to 70 (in conditions 3) (1)
- (higher rate of photosynthesis so) more glucose / carbohydrate is made for growth (1)

(c) **[2]**

- use (artificial) lights / lamps in the glasshouse (1)
- light intensity is the limiting factor (because the rate falls from 70 to 25 when the light is low even though temperature and carbon dioxide are the same) (1)

> **REJECT** adding more carbon dioxide or heating as the suggestion

Worked solution

(a) Compare conditions 1 and 2: only the temperature changes (8 °C to 18 °C) and the rate rises from 20 to 45. So in conditions 1 the low temperature is the limiting factor.

(b) The heater warms the air to about 18 °C, removing temperature as the limiting factor. It also adds carbon dioxide, so the conditions become like conditions 3, where the rate is 70 arbitrary units, compared with 20 in conditions 1. A higher rate of photosynthesis makes more glucose, which the lettuces use for growth, so they grow faster.

(c) In conditions 4, the temperature and carbon dioxide are the same as in conditions 3 but the rate is only 25, because the light intensity is low. Light intensity is the limiting factor, so lamps should be used to increase the light intensity.

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20 · Part 1 Core Leaf structure [7]

Mark scheme

(a) **[1]**

- (more surface for) absorbing more light / more light can be absorbed (1)

ALLOW more surface for the uptake / diffusion of carbon dioxide

(b) **[4]**

- (i) upper epidermis: lets light pass through to the mesophyll / palisade cells (1)
- (ii) palisade mesophyll: more chlorophyll so more light can be absorbed / more energy from light is transferred (1)
- (iii) spongy mesophyll: the air spaces allow carbon dioxide to diffuse to the (mesophyll) cells (1)
- (iv) stomata: let carbon dioxide diffuse into the leaf (1)

ALLOW (iii) and (iv) gas exchange / oxygen diffuses out, only if carbon dioxide is also mentioned

(c) **[2]**

- xylem brings water (a raw material for photosynthesis) to the leaf (1)
- phloem transports sucrose (made from the products of photosynthesis) away from the leaf (1)

ALLOW xylem brings mineral ions such as magnesium ions (for chlorophyll)

REJECT xylem carries sucrose / phloem carries water

Worked solution

(a) A large surface area means the leaf can absorb more light (and take in more carbon dioxide), so more photosynthesis can take place.

(b) (i) The upper epidermis is transparent, so light passes straight through to the photosynthesising cells below. (ii) Palisade mesophyll cells are packed with chloroplasts, which contain chlorophyll, so they absorb a lot of light. (iii) The air spaces in the spongy mesophyll let carbon dioxide diffuse quickly to the cells that photosynthesise. (iv) Stomata are pores that let carbon dioxide diffuse into the leaf (and oxygen diffuse out).

(c) Xylem carries water, one of the raw materials, to the leaf. Phloem carries sucrose made from the glucose of photosynthesis away from the leaf to the rest of the plant.

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